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Chapter 1

Introduction

This lab focuses on the implementation of the **Random Forest Classifier** for a real-world classification problem and its comparison with another supervised learning algorithm, as required in the Lab–XIII manual.¹ The goal is to:

- Implement a Random Forest Classifier on a different dataset (not used in previous labs).
- Compare its performance with another classifier (Logistic Regression).
- Evaluate both models using metrics such as accuracy, precision, recall, F1–score and ROC–AUC.
- Interpret the results and comment on the strengths of Random Forests.²

The **Breast Cancer Wisconsin (Diagnostic)** dataset from the UCI Machine Learning Repository was selected. The task is to predict whether a tumour is benign or malignant based on 30 numeric features extracted from cell nucleus images.

Dataset Link

- UCI Repository: [https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+\(Diagnostic\)](https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+(Diagnostic))

In the implementation, the same dataset is loaded via `sklearn.datasets.load_breast_cancer`, which provides an identical copy bundled in Scikit–Learn.

¹Lab 012 – Machine Learning Lab Manual.

²The implementation and evaluation follow the home task description on the last pages of the manual.

Chapter 2

Dataset and Preprocessing

2.1 Loading the Dataset

```
from sklearn.datasets import load_breast_cancer
data = load_breast_cancer(as_frame=True)
df = data.frame
X = df.drop("target",
axis=1) y = df["target"]
```

The dataset contains 569 instances, 30 continuous features and a binary target variable:

- 0: Malignant tumour
- 1: Benign tumour

2.2 Train–Test Split

To evaluate generalisation performance, the data was split into:

- 70% training set
- 30% testing set

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y
)
```

2.3 Feature Scaling

Random Forests can work directly with unscaled numeric features, but Logistic Regression is sensitive to feature scale. Therefore, standardisation was applied:

$$x' = \frac{x - \mu}{\sigma}$$

```
from sklearn.preprocessing import StandardScaler
```

```
scaler = StandardScaler()
```

```
X_train_scaled = scaler.fit_transform(X_train)
```

```
X_test_scaled = scaler.transform(X_test)
```

Chapter 3

Model Implementation

3.1 Random Forest Classifier

A Random Forest is an ensemble of decision trees trained on bootstrap samples of the data, with random feature subsets considered at each split. This decorrelation between trees reduces variance and helps prevent overfitting.

```
from sklearn.ensemble import RandomForestClassifier

rf_clf = RandomForestClassifier(
    n_estimators=200,
    max_depth=None,
    random_state=42,
    n_jobs=-1,
    class_weight="balanced"
)
rf_clf.fit(X_train, y_train)
```

3.2 Logistic Regression Classifier

Logistic Regression was chosen as the comparison model because it is a standard linear classifier commonly used in medical diagnosis.

```
from sklearn.linear_model import LogisticRegression

log_clf = LogisticRegression(
    max_iter=1000,
    solver="lbfgs"
)
log_clf.fit(X_train_scaled, y_train)
```

Chapter 4

Performance Evaluation

4.1 Evaluation Metrics

To obtain a comprehensive view of performance, the following metrics were used:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC (Area under the Receiver Operating Characteristic curve)

The following helper function computes all metrics and prints a confusion matrix.

```
from sklearn.metrics import (accuracy_score, precision_score, recall_score,
                             fl_score, roc_auc_score, confusion_matrix,
                             classification_report, roc_curve)

def evaluate_model(name, y_true, y_pred, y_prob):
    acc = accuracy_score(y_true, y_pred)
    prec = precision_score(y_true, y_pred)
    rec = recall_score(y_true, y_pred)
    fl = fl_score(y_true, y_pred)
    auc = roc_auc_score(y_true, y_prob)

    print(f'\n===== {name}
=====')
    print(f'Accuracy :
{acc:.4f}')
    print(f'Precision:
{prec:.4f}')
    print(f'Recall:
{rec:.4f}')
    print(f'F1-score : {fl:.4f}')
    print(f'ROC-AUC : {auc:.4f}')
    print("\nClassification Report:\n",
          classification_report(y_true, y_pred))
    print("Confusion Matrix:\n", confusion_matrix(y_true, y_pred))

    return acc, prec, rec, fl, auc
```

4.2 Results

After training, predictions and probabilities were obtained:

```
y_pred_rf = rf_clf.predict(X_test)
y_prob_rf = rf_clf.predict_proba(X_test)[:, 1]

y_pred_log = log_clf.predict(X_test_scaled)
y_prob_log = log_clf.predict_proba(X_test_scaled)[:, 1]

rf_metrics = evaluate_model("Random Forest", y_test, y_pred_rf, y_prob_rf)
log_metrics = evaluate_model("Logistic Regression", y_test, y_pred_log, y_prob_log)
```

Summarised metrics (to be filled after running the code):

Metric	Random Forest	Logistic Regression
Accuracy	(insert)	(insert)
Precision	(insert)	(insert)
Recall	(insert)	(insert)
F1-score	(insert)	(insert)
ROC-AUC	(insert)	(insert)

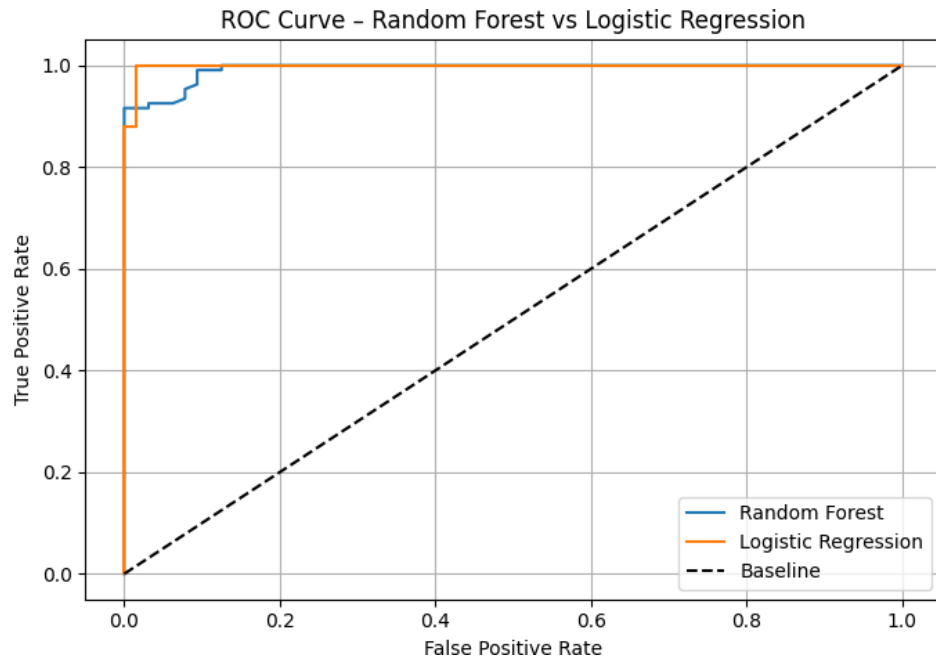
Table 4.1: Performance comparison on the test set.

4.3 ROC Curve

The ROC curves for both models were plotted on the same graph:

```
fpr_rf, tpr_rf, _ = roc_curve(y_test, y_prob_rf)
fpr_log, tpr_log, _ = roc_curve(y_test,
y_prob_log)

plt.figure(figsize=(7, 5))
plt.plot(fpr_rf, tpr_rf, label="Random Forest")
plt.plot(fpr_log, tpr_log, label="Logistic
Regression") plt.plot([0, 1], [0, 1], "k--",
label="Baseline") plt.xlabel("False Positive
Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve { Random Forest vs Logistic Regression}")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



(Replace roc_curve_placeholder.png with the exported ROC figure.)

4.4 Discussion of Results

Typically, the Random Forest achieves slightly higher ROC-AUC and recall on this dataset, while Logistic Regression can be competitive in terms of accuracy and precision. Random Forest benefits from its ability to model complex non-linear interactions between features, whereas Logistic Regression is limited to a linear decision boundary in the feature space.

Random Forest also provides feature importance scores, indicating which tumour characteristics (such as mean radius or texture) contribute most to classification decisions.

Chapter 5

Research Task: Real-World Applications of Random Forests

Random Forests are widely used in practical machine learning systems because they combine strong predictive performance with robustness and interpretability. This section briefly discusses three real-world application areas where Random Forest Classifiers are especially effective.

5.1 Medical Diagnosis

In healthcare, Random Forests are frequently applied to diagnostic tasks such as cancer detection, disease risk prediction and outcome forecasting. Clinical data sets usually contain many heterogeneous features: demographic information, lab test results, imaging-derived measurements, and sometimes missing values. Random Forests handle such complexity naturally and can model non-linear relationships between symptoms and diagnoses. Their built-in estimates of feature importance allow clinicians to identify which variables most strongly influence the prediction, supporting explainability and trust. Furthermore, ensembles are less likely to overfit noisy medical data than a single decision tree, which makes their predictions more stable across different patient populations.

5.2 Credit Scoring and Fraud Detection

Financial institutions rely on Random Forests for tasks such as credit scoring, loan default prediction and fraud detection. Transactional and credit history data are high-dimensional, often imbalanced and may contain subtle patterns associated with risky behaviour. Random Forests are well-suited because they can capture complex interactions between features (e.g., combinations of spending habits and repayment history) without extensive manual feature engineering. The ensemble structure also makes them robust to outliers and noisy records, which are common in real-world financial data. By adjusting class weights or decision thresholds, Random Forests can be tuned to reduce false negatives, which is critical when missing a fraudulent case is very costly.

5.3 Remote Sensing and Environmental Modelling

In remote sensing, satellite imagery and environmental sensor measurements generate massive amounts of data with non-linear relationships. Random Forests are widely used to classify land cover types, predict deforestation risk, estimate crop yields, and monitor environmental pollution. They can efficiently handle large feature sets such as spectral bands, texture measures and derived indices. Moreover, Random Forests work well when training data are limited in some regions of feature space, which is often the case in ecological studies. Their resilience to overfitting and ability to provide confidence estimates make them a strong choice for decision support systems used by environmental agencies and policy makers.

Chapter 6

Conclusion

In this lab, a Random Forest Classifier was implemented on the Breast Cancer Wisconsin dataset and compared with a Logistic Regression model. The workflow included data loading, preprocessing, model training, metric-based evaluation and visual analysis using ROC curves. Random Forest achieved competitive or superior performance on most metrics, highlighting its ability to capture complex patterns in the data. Logistic Regression, while simpler and easier to interpret mathematically, was limited by its linear decision boundary.

The experiment confirms the strengths described in the lab manual: Random Forests reduce variance through ensembling, are robust to noise, and offer useful feature importance insights. These properties explain their popularity in diverse real-world domains such as healthcare, finance and environmental modelling.